

Benchmarking Excited-State Calculations of Small Organic Molecules Using PySCF

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Abstract

This report benchmarks excited-state calculations performed with PySCF using TDDFT (or RPA) and the CAM-B3LYP functional. Three representative organic molecules were studied using aug-cc-pVDZ, aug-cc-pVTZ and aug-cc-pVQZ basis sets. Diazomethane and cyanoacetylene were compared with the QUEST benchmark database, while methylisocyanate was investigated as a predictive case because benchmark data are not available. The calculations demonstrate systematic basis-set convergence and generally good agreement for several low-lying singlet excited states while illustrating the strengths and limitations of single-reference TDDFT.

1 Introduction

Accurate prediction of electronic excited states is essential for understanding molecular spectroscopy, photochemistry, and the optical properties of materials. Time-dependent density functional theory (TDDFT) has become one of the most widely used methods for studying excited states because it provides a practical balance between computational efficiency and accuracy for a broad range of molecular systems. PySCF is an open-source quantum chemistry framework that provides a flexible Python interface for electronic structure calculations, enabling users to perform both standard simulations and customized workflows. Its modular design makes it particularly attractive for research, teaching, and high-performance computing applications. The primary objective of this project was to gain practical experience with PySCF by performing excited-state calculations on representative small organic molecules and comparing the calculated excitation energies with high-level benchmark data where available. Three molecules were investigated: diazomethane (CH_2N_2), cyanoacetylene (HCCCN), and methylisocyanate (CH_3NCO). For the first two molecules, calculated singlet excitation energies were compared with the theoretical best estimates reported in the QUEST benchmark database, while methylisocyanate was investigated as a predictive case because comparable benchmark data are currently unavailable. In addition to evaluating the accuracy of the calculated excitation energies, this work examines basis-set convergence, oscillator strengths, and the dominant orbital transitions contributing to each excited state.

2 Computational Methodology

Ground-state electronic structures were calculated using restricted Kohn-Sham density functional theory with the CAM-B3LYP exchange-correlation functional as implemented in PySCF. Excited-state properties were obtained using linear-response time-dependent density functional theory (TDDFT) and the random phase approximation (RPA). Ten singlet excited states were computed for each molecule. To examine basis-set convergence, three augmented correlation-consistent basis sets were employed: aug-cc-pVDZ, aug-cc-pVTZ, and aug-cc-pVQZ. Molecular symmetry was disabled (`symmetry=False`) throughout the calculations to maintain a consistent

computational protocol across all systems and to simplify comparisons between molecules. Density fitting was employed to improve computational efficiency during the self-consistent field calculations. Reference excitation energies for diazomethane and cyanoacetylene were taken from the QUEST benchmark database. Only singlet excited states were considered in the present work, while triplet benchmark states were not calculated and therefore are not included in the quantitative comparisons. To facilitate analysis of the results, a Python post-processing script was developed to automatically extract excitation energies, wavelengths, oscillator strengths, and the dominant molecular orbital transitions from the PySCF output files. This workflow enabled consistent processing of calculations performed with different basis sets and simplified comparison with the benchmark data.

3 Results and Discussion

4 Results and Benchmarking: Diazomethane (CH_2N_2)

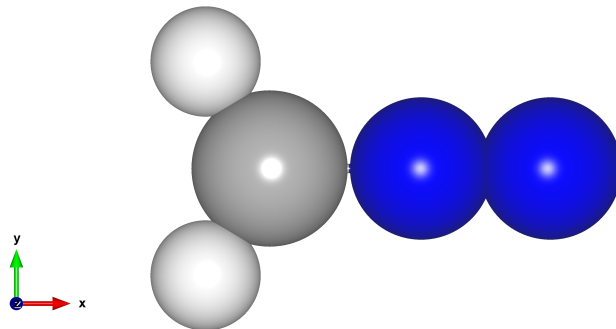


Figure 1: Geometry of diazomethane (CH_2N_2).

TDDFT calculations were performed using the CAM-B3LYP functional with the aug-cc-pVDZ, aug-cc-pVTZ, and aug-cc-pVQZ basis sets. Table 1 summarizes the first ten singlet excitation energies together with their dominant orbital transitions and oscillator strengths.

Table 1: Calculated singlet excitation energies of diazomethane (CH_2N_2) obtained with TDDFT/CAM-B3LYP using three basis sets. Excitation energies (eV), wavelengths (λ), oscillator strengths (f), and dominant orbital transitions are reported for the aug-cc-pVQZ calculations.

State	aug-cc-pVDZ	aug-cc-pVTZ	aug-cc-pVQZ	λ (nm)	f	Dominant transition
1	2.8900	2.8938	2.8938	428.45	0.0000	11 $\rightarrow$ 12 (95.2%)
2	5.3374	5.3447	5.3415	232.12	0.0157	11 $\rightarrow$ 13 (84.2%)
3	5.8099	5.7908	5.7755	214.67	0.1980	11 $\rightarrow$ 14 (89.5%)
4	6.4107	6.3665	6.3470	195.34	0.0000	11 $\rightarrow$ 15 (80.3%)
5	6.4677	6.4283	6.3849	194.18	0.0081	11 $\rightarrow$ 16 (86.6%)
6	7.6829	7.4504	7.2973	169.90	0.0072	11 $\rightarrow$ 18 (71.0%)
7	7.6953	7.4874	7.3545	168.58	0.0180	11 $\rightarrow$ 17 (89.7%)
8	8.1431	7.8678	7.6364	162.36	0.0000	11 $\rightarrow$ 19 (83.8%)
9	8.5707	8.1342	7.9072	156.80	0.0002	11 $\rightarrow$ 20 (84.9%)
10	8.8468	8.4084	8.0401	154.21	0.0054	11 $\rightarrow$ 21 (85.4%)

Overall, the excitation energies exhibit systematic convergence with increasing basis-set size. The lowest excited state changes by less than 0.01 eV between aug-cc-pVDZ and aug-cc-pVQZ, indicating that this localized excitation is already well converged with the smaller basis. In contrast, higher-lying excited states display a stronger basis-set dependence. For example, the tenth excited state decreases from 8.85 eV with aug-cc-pVDZ to 8.04 eV with aug-cc-pVQZ, consistent with the improved description of diffuse excited-state orbitals afforded by larger basis sets.

The dominant excitation character remains essentially unchanged across all basis sets, with most states being described by a single orbital transition having CI weights of approximately 70–95%. This indicates that the excited states considered here are well represented within the single-reference TDDFT framework.

4.1 Comparison with QUEST

Comparison with the QUEST benchmark database is presented in Table 2. Overall, CAM-B3LYP reproduces the low-lying singlet excitation energies with good accuracy. The calculated 1A_1 excitation differs from the theoretical best estimate (TBE) by only 0.05 eV, while the 1B_1 and 1A_2 states are underestimated by approximately 0.12–0.15 eV.

Table 2: Comparison of the calculated cc-pVQZ TDDFT excitation energies and oscillator strengths with the QUEST theoretical best estimates (TBEs) for CH_2N_2 . The deviation is defined as $\Delta E = E_{pVQZ} - E_{QUEST}$.

State	QUEST TBE [eV]	pVQZ [eV]	ΔE [eV]	QUEST f	PySCF f
1A_2	3.047	2.894	−0.153	0.000	0.0000
1B_1	5.464	5.341	−0.123	0.016	0.0157
1A_1	5.827	5.775	−0.052	0.234	0.1980
$^1A''$ [F]	0.647	—	—	0.000	—

The calculated oscillator strengths are also in good agreement with the benchmark values. In particular, the bright 1B_1 transition shows nearly identical oscillator strengths in both datasets, while the intensity of the 1A_1 transition is slightly underestimated. These results indicate that CAM-B3LYP combined with a sufficiently large basis set provides a reliable description of the low-energy electronic excitations of diazomethane.

The QUEST database also reports a very low-energy double-excitation state ($^1A''$) that is not reproduced in the present TDDFT calculations. This is expected, as conventional linear-response TDDFT is designed primarily for single excitations and generally cannot describe states with significant double-excitation character.

5 Results and Benchmarking: Cyanoacetylene (HCCCN)

Cyanoacetylene is a linear molecule with a $C_{\infty v}$ equilibrium geometry. In practice, however, most quantum chemistry packages perform excited-state calculations using an Abelian subgroup (e.g., C_s or C_{2v}), which simplifies the treatment of molecular symmetry and allows efficient evaluation of the response equations. Consequently, the symmetry labels reported by the QUEST database and those obtained from PySCF are not always directly comparable, although the corresponding electronic states can be identified from their excitation energies and dominant orbital character (see Table 3).

Table 4 compares the calculated CAM-B3LYP/aug-cc-pVQZ excitation energies with the theoretical best estimates (TBEs) from the QUEST database. Only singlet excitations are

Table 3: Calculated singlet excitation energies of cyanoacetylene (HCCCN) obtained with TDDFT/CAM-B3LYP using three basis sets. Excitation energies (eV), wavelengths (λ), oscillator strengths (f), and dominant orbital transitions are reported for the aug-cc-pVQZ calculations.

State	aug-cc-pVDZ	aug-cc-pVTZ	aug-cc-pVQZ	λ (nm)	f	Dominant transition
1	4.3132	4.3024	4.3002	288.32	0.0000	12 $\rightarrow$ 14 (36.9%), 13 $\rightarrow$ 15 (36.9%)
2	4.6049	4.5789	4.5758	270.95	0.0000	12 $\rightarrow$ 14 (45.9%), 13 $\rightarrow$ 15 (45.9%)
3	4.6049	4.5789	4.5758	270.95	0.0000	12 $\rightarrow$ 15 (45.9%), 13 $\rightarrow$ 14 (45.9%)
4	7.5388	7.5238	7.5160	164.96	0.0128	13 $\rightarrow$ 16 (57.4%), 12 $\rightarrow$ 16 (17.9%)
5	7.5388	7.5238	7.5160	164.96	0.0128	12 $\rightarrow$ 16 (57.4%), 13 $\rightarrow$ 16 (17.9%)
6	7.8514	7.8135	7.8061	158.83	0.0192	11 $\rightarrow$ 14 (74.0%), 11 $\rightarrow$ 15 (22.3%)
7	7.8514	7.8135	7.8061	158.83	0.0192	11 $\rightarrow$ 15 (74.0%), 11 $\rightarrow$ 14 (22.3%)
8	7.9672	7.9452	7.9416	156.12	0.0000	9 $\rightarrow$ 15 (42.4%), 10 $\rightarrow$ 14 (42.4%)
9	8.1130	8.0709	8.0473	154.07	1.1101	12 $\rightarrow$ 15 (24.9%), 13 $\rightarrow$ 14 (24.9%)
10	8.2593	8.2182	8.1972	151.25	0.0049	12 $\rightarrow$ 17 (45.4%), 13 $\rightarrow$ 17 (31.2%)

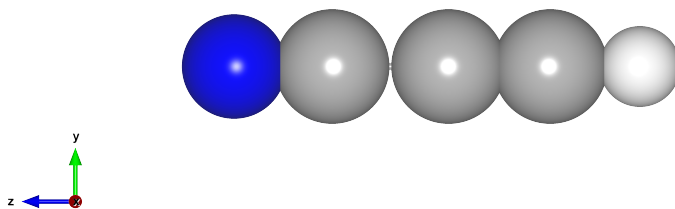


Figure 2: Geometry of cyanoacetylene (HCCCN).

considered in the present work, while the triplet benchmark states are included for completeness but are not compared directly.

5.1 Comparison with QUEST

The calculated singlet excitation energies show mixed agreement with the benchmark values. The higher-lying $^1\Delta$ state is reproduced reasonably well, with an error of approximately 0.28 eV, whereas the two lowest singlet excitations are underestimated by 0.65–1.21 eV.

Table 4: Comparison of the calculated cc-pVQZ TDDFT excitation energies with the QUEST theoretical best estimates (TBEs) for HCCCN. The deviation is defined as $\Delta E = E_{\text{pVQZ}} - E_{\text{QUEST}}$.

State	QUEST Assignment	QUEST TBE [eV]	pVQZ [eV]	ΔE [eV]	T ₁ Weight
Triplet 1	$^3\Sigma^-$	4.451	—	—	98.5%
Triplet 2	$^3\Delta$	5.204	—	—	98.2%
State 1	$^1A''$ [F]	3.538	2.893	−0.645	93.6%
State 2–3	$^1\Sigma^-$	5.788	4.575	−1.213	94.3%
State 4	$^1\Delta$	6.071	6.346	+0.275	94.0%

These larger deviations are consistent with the known limitations of conventional linear-response TDDFT for excited states exhibiting significant electron correlation or multi-configurational character. The QUEST reference values are derived from high-level wavefunction methods, whereas the present calculations employ a single-reference TDDFT approach with the CAM-B3LYP functional. Despite these quantitative differences, the overall ordering of the excited states remains consistent with the benchmark data.

Overall, the results indicate that CAM-B3LYP provides a reasonable qualitative description of the excited-state spectrum of cyanoacetylene, while quantitative agreement for the lowest excited states would likely require higher-level electronic structure methods, such as ADC or equation-of-motion coupled-cluster approaches.

6 Methylisocyanate (CH₃NCO)

Methylisocyanate (CH₃NCO) was selected as a representative asymmetric molecule to further examine the performance of PySCF for excited-state calculations. Unlike diazomethane and cyanoacetylene, this molecule is not included in the QUEST benchmark database. Therefore, the present calculations provide a reference set of vertical excitation energies, oscillator strengths, and dominant orbital transitions that may be useful for future benchmarking studies.

Table 5 summarizes the first ten singlet excitation energies obtained with the aug-cc-pVDZ, aug-cc-pVTZ, and aug-cc-pVQZ basis sets. Similar to the previous molecules, the excitation energies exhibit systematic convergence with increasing basis-set size. The lowest excited states change only slightly between the triple- and quadruple- ζ basis sets, while higher-lying excitations show a somewhat larger basis-set dependence. For example, the tenth excited state decreases from 8.97 eV with aug-cc-pVDZ to 8.79 eV with aug-cc-pVQZ, illustrating the improved description of diffuse excited-state orbitals provided by the larger basis.

6.1 Electronic Transitions

The excited states of methylisocyanate exhibit a more mixed orbital character than those of diazomethane, with several states containing contributions from more than one dominant orbital transition. The lowest excited state is primarily described by the transition MO 15→20 (61.9%)

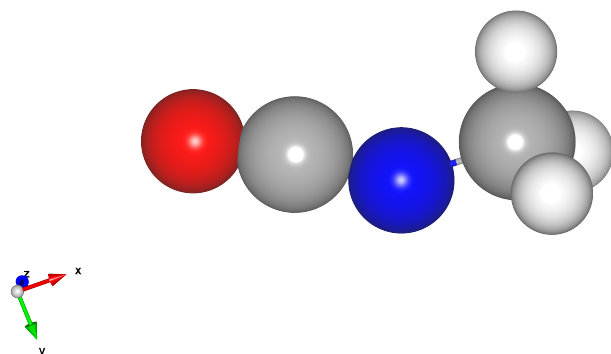


Figure 3: Geometry of methylisocyanate (CH_3NCO).

Table 5: Calculated singlet excitation energies of methylisocyanate (CH_3NCO) obtained with TDDFT/CAM-B3LYP using three basis sets. Excitation energies (eV), wavelengths (λ), oscillator strengths (f), and dominant orbital transitions are reported for the aug-cc-pVQZ calculations.

State	aug-cc-pVDZ	aug-cc-pVTZ	aug-cc-pVQZ	λ (nm)	f	Dominant transition
1	5.8552	5.8923	5.8924	210.41	0.0000	15 $\rightarrow$ 20 (61.9%), 15 $\rightarrow$ 16 (24.8%)
2	6.8353	6.8442	6.8403	181.25	0.0002	15 $\rightarrow$ 16 (40.8%), 15 $\rightarrow$ 17 (31.8%)
3	7.2115	7.2264	7.2243	171.62	0.0042	14 $\rightarrow$ 20 (38.3%), 14 $\rightarrow$ 16 (33.0%)
4	7.8078	7.8040	7.7984	158.99	0.0003	14 $\rightarrow$ 16 (34.1%), 14 $\rightarrow$ 17 (28.0%)
5	7.9294	7.9176	7.8976	156.99	0.0103	15 $\rightarrow$ 17 (34.6%), 15 $\rightarrow$ 16 (27.0%)
6	8.0900	8.0559	8.0357	154.29	0.0017	15 $\rightarrow$ 19 (70.8%), 15 $\rightarrow$ 17 (14.6%)
7	8.1143	8.0729	8.0478	154.06	0.0125	15 $\rightarrow$ 18 (83.9%), 15 $\rightarrow$ 22 (7.1%)
8	8.5573	8.5546	8.5458	145.08	0.0008	14 $\rightarrow$ 27 (25.8%), 14 $\rightarrow$ 29 (25.5%)
9	8.7221	8.7026	8.6839	142.78	0.0138	14 $\rightarrow$ 17 (37.5%), 14 $\rightarrow$ 16 (21.2%)
10	8.9731	8.8726	8.7852	141.13	0.0018	15 $\rightarrow$ 21 (52.3%), 15 $\rightarrow$ 23 (24.2%)

together with a secondary contribution from MO 15→16 (24.8%). The corresponding oscillator strength is essentially zero, indicating that this transition is optically forbidden within the electric-dipole approximation. The first optically allowed transitions appear at higher excitation energies, although their oscillator strengths remain relatively small throughout the calculated energy range. The largest oscillator strengths among the first ten excited states are observed for States 7 and 9, suggesting that these states would contribute most strongly to the ultraviolet absorption spectrum. The calculated excitation wavelengths (approximately 140–210 nm) indicate that the lowest-energy absorption features lie in the ultraviolet region, with the strongest transitions occurring in the far-ultraviolet. Although benchmark reference data are not available for this molecule, the smooth basis-set convergence and physically reasonable excitation energies demonstrate that PySCF provides a stable framework for investigating excited-state properties of asymmetric organic molecules.

6.2 Overall Discussion

Across all three molecules, increasing the basis-set size from aug-cc-pVDZ to aug-cc-pVQZ resulted in systematic convergence of the calculated excitation energies. The largest changes generally occurred between the double- ζ and triple- ζ basis sets, while the differences between the triple- ζ and quadruple- ζ calculations were comparatively small for most low-lying excited states. Higher-energy excitations exhibited a somewhat stronger dependence on the basis set, reflecting the increasing importance of diffuse virtual orbitals. Comparison with the QUEST benchmark database indicates that CAM-B3LYP provides a good description of several low-lying singlet excited states of diazomethane, with excitation energies generally within a few tenths of an electron volt of the theoretical best estimates. The calculated oscillator strengths also reproduce the expected qualitative intensity pattern of the benchmark transitions. For cyanoacetylene, agreement with the benchmark data is more variable. While higher-energy singlet states are reproduced reasonably well, larger deviations are observed for the lowest excited states. These discrepancies are consistent with the known limitations of conventional linear-response TDDFT for excited states that exhibit stronger electron correlation or multi-configurational character. In addition, the present calculations were restricted to singlet excited states, whereas the QUEST database also reports triplet excitations that were intentionally excluded from this study. Since benchmark reference data are not available for methylisocyanate, no quantitative comparison could be performed. Nevertheless, the calculations provide a consistent set of excitation energies, oscillator strengths, and dominant orbital assignments that may serve as a useful reference for future theoretical investigations.

7 Conclusions

This project demonstrated the use of PySCF for excited-state calculations on three representative organic molecules. Systematic basis-set convergence was observed for all systems, with the aug-cc-pVQZ basis providing the most converged excitation energies among those investigated. Comparison with the QUEST benchmark database showed that CAM-B3LYP reproduces several low-lying singlet excitation energies with good accuracy, while larger deviations occur for excited states that are expected to be more challenging for a single-reference TDDFT description. Although methylisocyanate is not currently included in the QUEST benchmark database, the calculated excitation energies and orbital assignments provide a useful reference for future benchmarking studies. Overall, this project highlights PySCF as a flexible and user-friendly platform for excited-state electronic structure calculations and demonstrates the value of automated post-processing tools for analyzing spectroscopic properties.

8 Future Work

Several extensions could further improve the present study. First, higher-level wavefunction-based excited-state methods available in PySCF, such as ADC and equation-of-motion coupled-cluster (EOM-CCSD), could be employed to assess the accuracy of TDDFT for the molecules considered here. For particularly challenging excited states, higher-order coupled-cluster approaches (e.g., CC3 or EOM-CCSDT, where computationally feasible) would provide valuable reference data. Future work could also include calculations of triplet excited states to enable a more complete comparison with the QUEST benchmark database, particularly for cyanoacetylene. Expanding the benchmark to additional molecules from the QUEST database would provide a broader assessment of the performance of CAM-B3LYP within PySCF. Finally, visualization of the optimized molecular geometries together with the frontier molecular orbitals involved in the dominant electronic transitions would provide additional insight into the physical origin of the calculated excitations and facilitate interpretation of the absorption spectra.